

KODIAK, AK, USA

WMO: 703500

Lat: 57.751N

Lon: 152.486W

Elev: 24

StdP: 101.03

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25501

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-10.2	-19.0	0.7	-10.1	-17.1	0.8	-8.8	16.1	1.1	14.7	1.0	5.2	290	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	21.0	14.4	18.9	13.3	17.3	12.8	15.2	19.4	14.2	17.4	13.5	16.3	4.5	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	9.6	16.0	12.7	9.2	15.0	12.1	8.8	14.4	42.3	19.4	39.7	17.5	37.9	16.2	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.3	10.9	DB	-15.5	24.5	3.1	2.0	-17.7	25.9	-19.5	27.1	-21.2	28.2	-23.5	29.7	(4)
(5)			WB	-16.1	17.0	3.0	1.5	-18.3	18.1	-20.0	19.0	-21.7	19.8	-23.9	20.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-0.6	0.1	0.2	3.5	7.2	10.4	12.9	13.3	10.1	5.4	1.7	-0.4
	DBStd	6.01	4.42	3.74	3.69	2.45	2.52	2.56	2.30	2.21	2.36	3.02	3.77	4.18
	HDD10.0	1963	327	276	303	196	94	23	1	1	26	144	250	322
	HDD18.3	4743	586	509	561	446	345	239	169	158	248	402	500	580
	CDD10.0	266	0	0	0	0	7	34	92	103	29	1	0	0
	CDD18.3	3	0	0	0	0	0	1	1	2	0	0	0	0
	CDH23.3	15	0	0	0	0	0	4	7	4	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	4.6	5.2	5.1	5.3	5.0	4.5	3.8	3.3	3.6	4.1	4.7	5.1	5.5
Precipitation	PrecAvg	1723	195	148	119	104	123	140	93	122	173	191	152	183
	PrecMax	2227	436	316	252	236	322	429	259	283	308	435	390	503
	PrecMin	972	6	37	21	7	25	18	21	16	30	80	17	30
	PrecStd	332	111	79	62	48	83	93	55	70	69	95	79	111
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	6.9	8.1	12.8	18.7	22.9	24.3	23.5	18.9	12.8	9.7	7.5
		MCWB	4.5	4.7	4.7	6.9	10.3	14.9	16.3	16.0	12.5	9.3	7.8	6.2
	2%	DB	5.9	5.8	6.7	9.8	15.6	19.1	20.9	20.7	16.5	11.5	8.6	6.6
		MCWB	5.0	4.3	4.5	5.9	9.1	12.7	14.8	14.4	11.8	8.8	7.3	5.6
	5%	DB	5.3	5.1	5.8	8.0	13.2	16.8	18.7	18.6	15.1	10.6	7.8	5.8
		MCWB	4.5	4.0	4.0	4.9	8.1	11.8	13.6	13.5	11.2	8.6	6.5	4.9
	10%	DB	4.7	4.4	4.8	7.0	11.3	14.7	16.9	17.1	13.8	9.9	7.1	5.0
		MCWB	3.9	3.3	3.3	4.6	7.4	10.8	13.2	13.0	10.8	8.0	5.7	4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	5.7	6.4	7.9	11.6	15.3	16.6	16.9	14.3	10.9	8.8	6.8
		MCDB	6.3	6.1	7.2	10.9	16.8	22.1	22.7	22.0	16.5	11.3	9.4	7.2
	2%	WB	5.4	4.9	5.3	6.6	9.8	13.5	15.2	15.3	12.9	10.1	7.9	5.9
		MCDB	5.6	5.3	6.1	8.6	14.3	18.1	20.0	18.8	15.0	10.8	8.4	6.4
	5%	WB	4.8	4.2	4.5	5.8	8.9	12.2	14.2	14.4	12.0	9.2	6.9	5.1
		MCDB	5.1	4.8	5.3	7.3	12.3	15.5	17.5	17.2	13.9	10.1	7.7	5.7
	10%	WB	4.1	3.6	3.6	5.2	8.0	11.2	13.5	13.7	11.4	8.5	6.0	4.3
		MCDB	4.5	4.2	4.5	6.5	10.6	13.8	16.3	16.1	13.0	9.6	6.9	4.8
Mean Daily Temperature Range	5% DB	MDBR	4.6	4.9	5.3	5.4	5.7	5.5	5.4	6.0	6.2	6.0	5.3	5.0
		MCDBR	3.7	4.9	5.9	7.2	9.7	10.2	9.4	8.8	8.2	6.4	4.9	4.1
		MCWBR	3.4	4.2	4.5	5.0	5.4	5.5	4.9	4.6	4.8	4.8	4.4	3.9
	5% WB	MCDBR	3.1	4.0	4.6	5.6	7.8	8.5	8.0	7.2	6.3	4.9	3.9	3.5
		MCWBR	3.2	3.9	4.0	4.3	4.7	4.9	4.6	4.3	4.4	4.4	4.0	3.7
Clear-Sky Solar Irradiance	taub	0.276	0.320	0.343	0.380	0.387	0.379	0.375	0.366	0.353	0.332	0.303	0.302	
	taud	2.338	2.300	2.265	2.218	2.281	2.398	2.457	2.485	2.464	2.364	2.190	2.052	
	Ebn at Noon	604	716	800	826	846	860	856	839	792	692	519	384	
	Edh at Noon	44	72	97	119	120	109	100	91	80	67	50	44	
All-Sky Solar Radiation	RadAvg	0.41	0.98	2.21	3.32	4.38	4.77	4.55	3.69	2.52	1.37	0.60	0.30	
	RadStd	0.05	0.14	0.31	0.37	0.48	0.49	0.41	0.40	0.32	0.15	0.08	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.77	+1.23	N/A	N/A	N/A	N/A	-197	-255	N/A	N/A
(47) Regional (4 neighbors)		+0.78	+1.33	+1.22	N/A	N/A	N/A	-197	-255	N/A	N/A

Nomenclature: See separate page